(1) $\sqrt[4]{81 x^{-8} z^4}$

Sol)

$$\left(3^4 x^{-8} z^4\right)^{\frac{1}{4}}$$

$$3^{4\times\frac{1}{4}} \quad x^{-8\times\frac{1}{4}} \quad z^{4\times\frac{1}{4}}$$

$$3^1 x^{-2} z$$

$$3 \cdot 1 \cdot z$$

$$x^2$$

$$\boxed{\frac{3z}{x^2}} \quad \text{Ans}$$

(2) $\sqrt[3]{256\, a^6 b^{12} c^9}$

Sol)

2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

$$\sqrt[3]{2\times2\times2\times2\times2\times2\times2\times2\times2\; a^6 b^{12} c^9}$$

$$\left(2^3 \times 2^3 \times 2^2\, a^6 b^{12} c^9\right)^{\frac{1}{3}}$$

$$\left(2^{3\times\frac{1}{3}} \times 2^{3\times\frac{1}{3}} \times 2^{2\times\frac{1}{3}} \times a^{6\times\frac{1}{3}} \times c^{9\times\frac{1}{3}} \times b^{12\times\frac{1}{3}}\right)$$

$$2\times2\times2^{\frac{2}{3}}\times a^2 \times c^3 \times b^4$$

$$\left(2^{\frac{2}{3}+1+1} \times a^2 + c^3 * b^4\right)$$

$$\left(2^{\frac{2+6}{3}} \times a^2 + c^3 \times b^4\right)$$

$$2^{\frac{8}{3}}$$

$$\boxed{4 \times \sqrt[3]{2^2} \times a^2 \times c^3 \times b^4}$$

Ans